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Laguna State Polytechnic University
Province of Laguna

Professional Installation Manual

Ubuntu Server Deployment for ABC Startup Solutions
Mini Project – BSIT 4A
System Administration and Maintenance – Week 03

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1. Introduction

This manual documents the complete process of deploying Ubuntu Server for ABC Startup Solutions' first Linux server, intended to support file sharing, remote administration, database hosting, web hosting, and internal services. It is written so that another system administrator, with no prior context, can reproduce this deployment from start to finish, verify the result, and troubleshoot common issues encountered along the way.

The manual covers installation using Oracle VirtualBox, initial post-install configuration, a full verification checklist, a troubleshooting reference built from real issues encountered during this deployment, and a set of best practices for future installations.

2. Ubuntu Server Installation Guide

2.1 Prerequisites

- Oracle VirtualBox (or VMware Workstation) installed on the host machine
- Ubuntu Server LTS ISO downloaded (latest stable release)
- Host machine with sufficient free RAM/CPU for the target VM allocation

2.2 Virtual Machine Specifications

Component	Recommended Value
Name	descriptive, e.g. Ubuntu-Server-Week03
RAM	Approx. 40–50% of host total RAM (minimum ~1.5 GB)
CPU	1–2 virtual processors
Network	40 GB (VDI, dynamically allocated)
Storage	NAT (or Bridged if instructed)
Optical Drive	Ubuntu Server ISO

2.3 Installation Steps

1. Create a new VM in VirtualBox with the specifications above.
2. Attach the Ubuntu Server ISO to the virtual optical drive and start the VM.
3. Select language (English) and keyboard layout (English US).
4. Choose “Ubuntu Server” as the installation base (not the minimized variant, unless a minimal footprint is required).
5. Accept DHCP networking, or configure static addressing if instructed. Record the assigned IP address.
6. Configure guided storage using the entire disk with LVM.
7. Enter the server's hostname, an administrator full name, a non-root username, and a strong password.
8. Enable “Install OpenSSH server” for remote administration.



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9. Skip additional server snaps unless specifically required.

10. Allow installation to complete, then remove the ISO from the virtual optical drive before rebooting.

```
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
Installing packages on target system: ['grub-pc']
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
updating initramfs configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
curtin command in-target
curtin command in-target
calculating extra packages to install
installing openssh-server
retrieving openssh-server
curtin command system-install
unpacking openssh-server
curtin command system-install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquityLate/run:

[ View full log ]
[ Reboot Now ]
```

3. Initial Server Configuration

After the first boot, log in with the administrator account created during installation. The welcome banner displays a snapshot of system status, including the assigned IP address, memory and disk usage, and available updates.

```

Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-137-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Aug 11 02:52:46 PM UTC 2026

System load:          0.33
Usage of /:            24.6% of 10.53GB
Memory usage:         13%
Swap usage:           0%
Processes:            100
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fedc:480e

Expanded Security Maintenance for Applications is not enabled.

34 updates can be applied immediately.
1 of these updates is a standard security update.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

syu@server01:~$
```

3.1 Apply System Updates

Update the package index and install available updates immediately after first login:

`sudo apt update`
`sudo apt upgrade -y`



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3.2 Confirm SSH Is Running

Enable and start the SSH service, then confirm it is active:

```
sudo systemctl enable ssh sudo systemctl start ssh systemctl status ssh
```

4. Server Verification Procedures

Use the following checklist to confirm the server is fully functional before handing it off or placing it into service.

4.1 Login Verification

Confirm the administrator account can log in with the configured username and password.

4.2 Hostname and IP Address

Confirm the hostname matches the intended server name, and that the network interface has a valid IP address.

4.3 Internet Connectivity

```
ping -c 4 google.com
```

A successful test shows 0% packet loss.

```
Ubuntu-Server-Week03 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

syy@server01:~$ hostname
server01
syy@server01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:dc:48:0e brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86072sec preferred_lft 86072sec
    inet6 fd17:625c:f037:2:a00:27ff:fedc:480e/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86074sec preferred_lft 14074sec
    inet6 fe80::a00:27ff:fedc:480e/64 scope link
        valid_lft forever preferred_lft forever
syy@server01:~$ ping -c 4 google.com
PING google.com (172.217.26.78) 56(64) bytes of data:
64 bytes from sin10s02-in-f14.1e100.net (172.217.26.78): icmp_seq=1 ttl=64 time=15.2 ms
64 bytes from sin10s02-in-f14.1e100.net (172.217.26.78): icmp_seq=2 ttl=64 time=19.3 ms
64 bytes from sin10s02-in-f14.1e100.net (172.217.26.78): icmp_seq=3 ttl=64 time=161 ms
64 bytes from sin10s02-in-f14.1e100.net (172.217.26.78): icmp_seq=4 ttl=64 time=13.4 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 13.392/52.349/161.488/63.048 ms
syy@server01:~$
```



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4.4 System Updates

Confirm apt update / apt upgrade complete without errors.

```
Installing new version of config file /etc/fuupd/remotes.d/lvls-testing.conf ...
Installing new version of config file /etc/fuupd/remotes.d/lvls.conf ...
Installing new version of config file /etc/fuupd/remotes.d/vendor-directory.conf ...
Installing new version of config file /etc/grub.d/35-fuupd ...
Installing new version of config file /etc/update-motd.d/35-fuupd ...
fuupd-refresh.service is a disabled or a static unit not running, not starting it.
fuupd.service is a disabled or a static unit not running, not starting it.
Setting up cloud-init (26.1-0ubuntu1~24.04.1) ...
Installing new version of config file /etc/cloud/templates/chrony.conf.freebsd.tpl ...
Setting up apport-core-dump-handler (2.20.3-0ubuntu0.1) ...
Setting up apport (2.20.3-0ubuntu0.1) ...
apport-autoreport.service is a disabled or a static unit not running, not starting it.
Processing triggers for dbus (1.14.10-4ubuntu1) ...
Processing triggers for udev (255.4-0ubuntu0.17) ...
Processing triggers for install-info (7.1-3build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.0) ...
Processing triggers for rsyslog (8.2312.0-3ubuntu9.3) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for initramfs-tools (0.142ubuntu25.0) ...
update-initramfs: Generating /boot/initrd.img-6.8.0-107-generic
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
/etc/needrestart/restart.d/systemd-manager
systemctl restart packagekit.service

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service

No containers need to be restarted.

User sessions running outdated binaries:
syu @ user manager service: systemd[1150]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
syu@server01:~$
```

4.5 SSH Service Status

Confirm the SSH service shows Active: active (running), not inactive or failed.

```
No containers need to be restarted.

User sessions running outdated binaries:
syu @ user manager service: systemd[1150]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
syu@server01:~$ systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
TriggeredBy: ● ssh.socket
           Docs: man:sshd(8)
                man:sshd_config(5)
syu@server01:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/ssh.service → /usr/lib/systemd/system/ssh.service.
syu@server01:~$ sudo systemctl start ssh
syu@server01:~$ systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-08-11 15:08:52 UTC; 45s ago
TriggeredBy: ● ssh.socket
           Docs: man:sshd(8)
                man:sshd_config(5)
   Process: 13648 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 13650 (sshd)
     Tasks: 1 (limit: 4650)
    Memory: 2.1M (peak: 2.4M)
       CPU: 114ms
   CGroup: /system.slice/ssh.service
           └─13650 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 11 15:08:52 server01 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 11 15:08:52 server01 sshd[13650]: Server listening on 0.0.0.0 port 22.
Aug 11 15:08:52 server01 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
syu@server01:~$
```



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5. Troubleshooting Guide

The following issues were encountered during this deployment and are documented here for future administrators facing similar symptoms.

Symptom	Likely Cause	Resolution
VirtualBox won't save VM settings; "Invalid settings detected", more than 80% of host memory assigned	Host machine has very limited RAM (e.g. 3.5 GB) and the configured VM memory exceeds a safe share of it	Lower Base Memory in Settings → System to roughly 40–50% of total host RAM (e.g. 1536–2048 MB)
Installer or running VM freezes; console shows "watchdog: BUG: soft lockup – CPU stuck"	Host CPU is overloaded by other running applications, starving the VM of processing time	Close unnecessary host applications/browser tabs before and during installation; wait several minutes for recovery before resetting
VM disappears from VirtualBox Manager; "Failed to open virtual machine ... same UUID as an existing virtual machine"	A stale or duplicate entry exists in VirtualBox's internal registry (VirtualBox.xml) after an unexpected shutdown	Close VirtualBox and background VBoxSVC processes, open %USERPROFILE%\VirtualBox\VirtualBox.xml, back it up, remove the conflicting <MachineEntry> line, then re-add the VM via File → Add Machine
systemctl status ssh shows "Active: inactive (dead)" even though OpenSSH was selected during installation	The service was installed but left socket-activated / disabled rather than actively running on first boot	Run sudo systemctl enable ssh followed by sudo systemctl start ssh, then re-check with systemctl status ssh
Installer screen appears to have no visible "Done" or continue button	The VirtualBox window/terminal resolution is too small to display the full installer screen	Maximize the VirtualBox window, or press Tab repeatedly to cycle focus to the off-screen button and press Enter
VM screen shows "Press Ctrl-Alt-Del to unlock" and normal key presses do nothing	The host OS intercepts the Ctrl-Alt-Del combination before it reaches the guest VM	Use the VM window's Input → Keyboard → Insert Ctrl-Alt-Del menu option (only available inside the running VM's own window, not the VirtualBox Manager)

6. Best Practices

- Match VM resource allocation to the host machine's actual specifications, not just the assignment's stated minimums — check host RAM/CPU in Task Manager before creating a VM.
- Prefer CLI-only / Server Core installations over GUI/Desktop Experience editions when working on limited hardware; a GUI consumes RAM and CPU that a server role does not need.



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- Always verify a service's actual running state with `systemctl status` (Linux) or equivalent tools after installation — do not assume a checked “install” option means the service is active.
- Record IP addresses, hostnames, and credentials immediately as they are assigned, rather than relying on memory or having to re-query the system later.
- Take a screenshot at each significant installation milestone as it happens; recreating documentation afterward is far slower and less accurate.
- Keep the installation ISO attached only until first boot completes, then remove it from the virtual optical drive to prevent the VM from booting into the installer again.
- Enable OpenSSH (or the platform's remote-management equivalent) during installation whenever remote administration is a project requirement, rather than adding it after the fact.
- Apply operating system updates (`apt update` & `apt upgrade`, or Windows Update) immediately after first login, before deploying any application workloads.
- Document problems and their solutions as they are resolved, while the exact cause and fix are still fresh, rather than trying to reconstruct the troubleshooting process later.